Is Coding *Dead?*

Java in the AI Era.

What's real, what's hype, and where you fit.

v1.5 - @frankgreco

Is coding dead?

Two kinds of noise drown out a calmer, more useful truth.

“Coding is dead”

Doomers say AI ends the job. Reality: the floor rose and boilerplate got cheap. But software is more than typing, and demand for people who understand systems isn't shrinking.

Doomer

Snake oil & YouTube-ism

Influencers say prompts replace engineers. They sell courses/seminars, not outcomes. Tune out the thumbnails; watch what actually ships.

Snake oil

AI is a real productivity shift not a replacement.

The winners pair it with real engineering. Java already runs the enterprise systems that AI needs to plug into.

The **value of AI** compounds with the **value of Java.**

Why Java matters in the AI era

AI doesn't run on notebooks in production. It runs where business applications already live. That's Java.

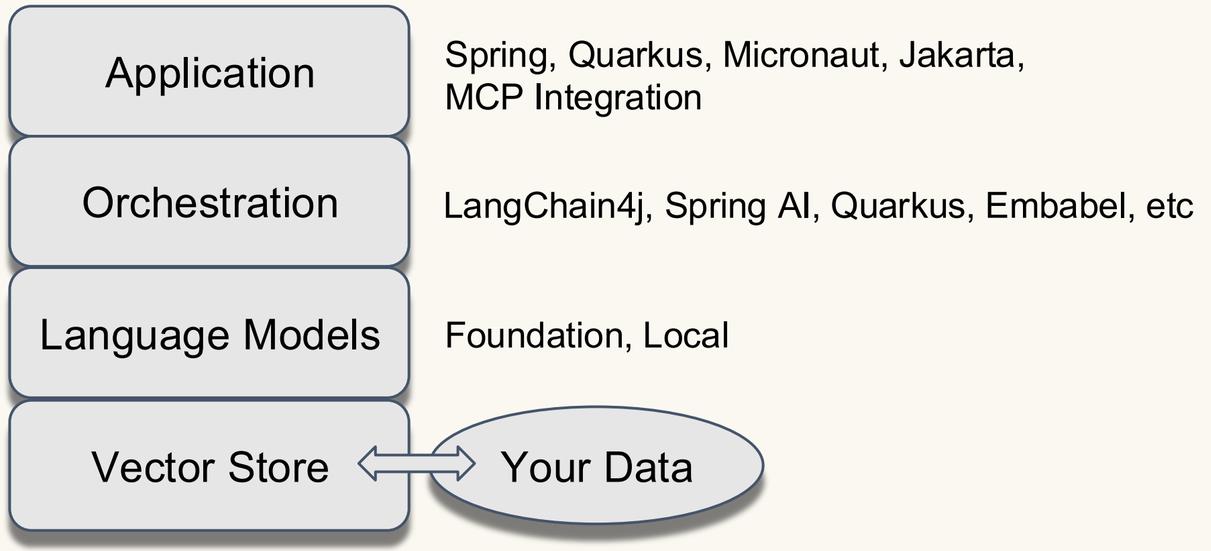
WHERE PYTHON FITS

Research, model training, data science, notebooks, the cutting edge of ML tooling.

WHERE JAVA FITS

Production & enterprise integration, concurrency, scale, type safety, ***security***, systems of record.

THE AI STACK FOR JAVA



AI concepts you must know

Start with fundamental concepts. How a model behaves and how to ground it in your data.

THE MODEL

How LLMs work

Non-deterministic by design. The same prompt can give different answers. You don't program it, you steer it.

Non-determinism | Prompting | Context | LLM vs Chatbot

YOUR DATA

RAG

Retrieval-Augmented Generation (RAG) grounds answers in your own content, so the model cites what you know, not what it guessed.

Similarity | Embeddings | Vector DBs

RAG FLOW

Question → Embed → Vector DB → Retrieve → LLM + context → Grounded answer

Agents

An LLM that can take actions. Looping until a goal is met turns answers into work.

WHAT MAKES IT AN AGENT

Give a reasoning model some tools and a goal, and let it decide the steps. It calls your functions to get things done.

Loop **Tools / function calling** **Memory**

WHEN TO REACH FOR ONE

Multi-step tasks that touch real systems… Query a DB, call an API, file a ticket. Note: for simpler Q&A over docs, modern RAG is enough.

THE AGENT LOOP

Plan → **Act · call a tool** → **Observe**

↻ repeat until the goal is met tools: query DB · hit API · run code

Your AI toolkit

Some suggestions for building generative AI apps without leaving the JVM.

01	**AI coding assistants**	In-IDE pair programming for everyday coding, refactors and tests.
02	**Local LLMs**	Run models on your own machine… private, offline, zero per-token cost. (Ollama/LMStudio)
03	**LangChain4j**[1]	The orchestration layer: chat, RAG, tools and agents, idiomatic Java.
04	**Quarkus LC4j & Spring AI**	First-class framework integration — config, DI and observability built in.
05	**Embabel**	Open-source agent framework based on Goal-Oriented Action Planning (GOAP)

1 

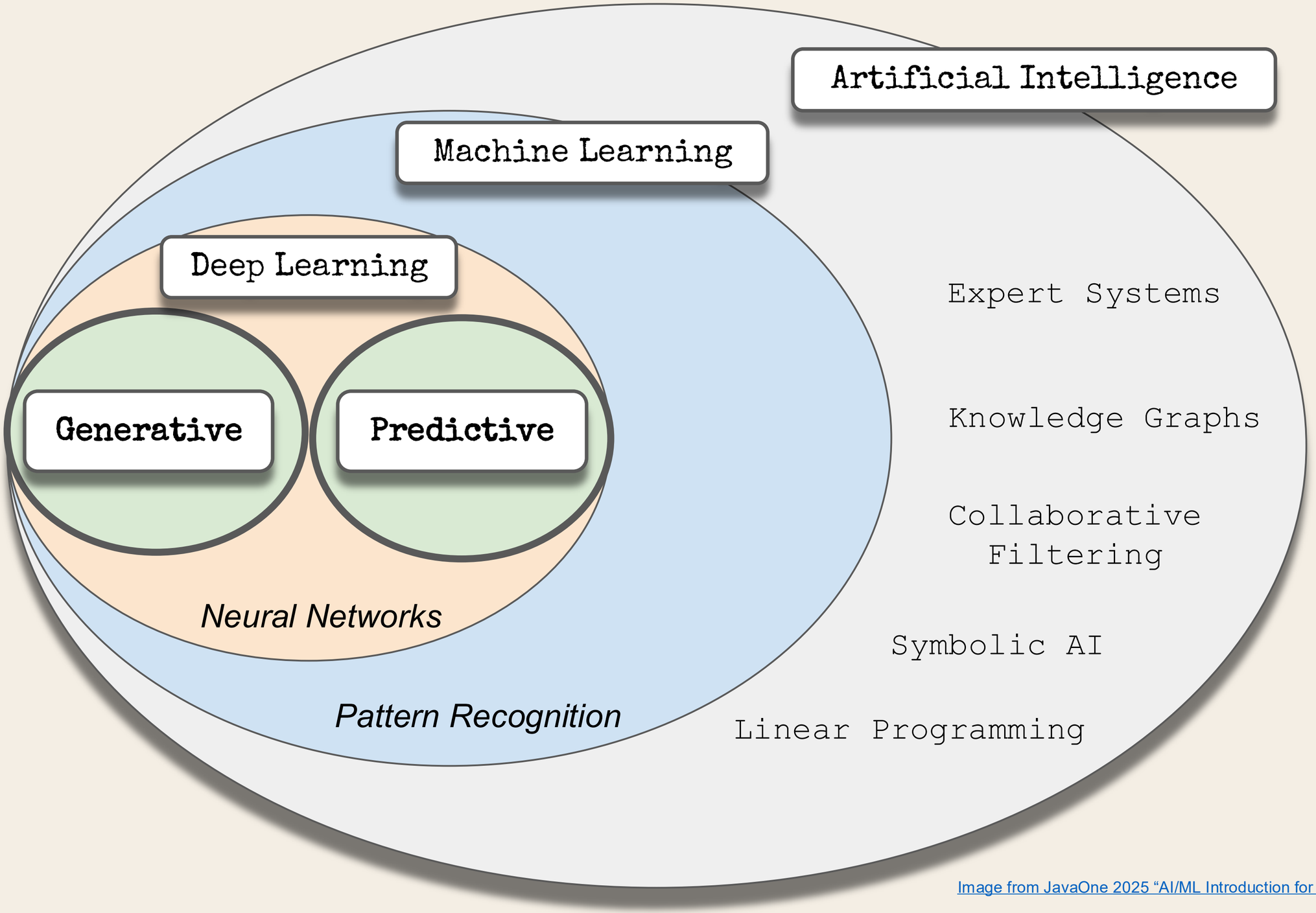


Image from JavaOne 2025 “AI/ML Introduction for Java Developers”

Predictive AI (vs Generative AI) on the JVM

Not everything is GenAI and LLMs. Classic machine learning still ships and it's all here in Java.

DeepLearning4j (Eclipse Deeplearning4j)

Distributed deep learning & neural nets, built for the JVM and big-data stacks.

DeepNetts

Pure-Java deep learning with a Java-native API. Very easy to embed straight into apps. See **JSR 381**

Weka

The classic ML workbench: classification, regression and clustering, out of the box.

Mahout

Scalable, distributed linear algebra and machine learning for large datasets.

Reach for these when the job is forecasting · classification · anomaly detection. No LLM required.

Where to focus your growth

The model writes code. Your edge is everything around the code.

THE COMPOUNDING SKILL

Critical thinking vs. code generation

Generating code is cheap. Knowing whether it's correct, safe and right for the problem is the job.

Heterogeneous system design

Java, Python, models & data working as one system.

Collaboration

Working with AI, and with people, as a force multiplier.

Security & privacy

Where data goes, prompt injection, what stays on-prem.

Ops & risk

Costs, observability, explainability, risk management.

Getting Started

Before you jump into the deep end of the pool, learn how to swim

Core Technical Skills

Modern Dev Skills

Hands-on Practice and Open Source

Career Readiness and Community

Practical projects for your CV

Small, shippable, and proof that you can build applications and not just prompt.

Simple RAG

Private Chatbot

Agent with Java tools

Streaming AI Response

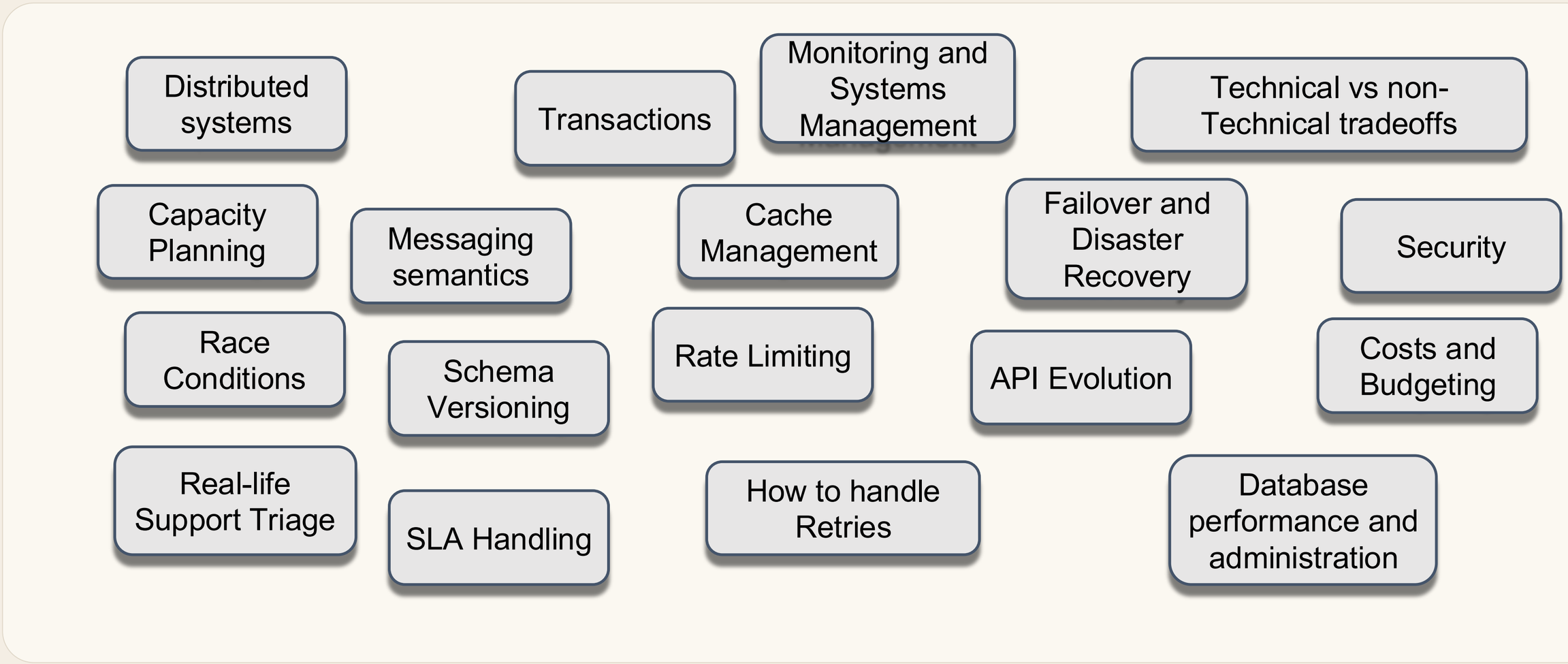
More Tips for New Developers

Consider these suggestions

- Problem Selection and Ambition
- Domain Expertise and Judgment
- Ownership, Execution, and Quality
- Learning, Experimentation, and Resilience
- Career, Relationships, and Responsible AI

Software Engineering is more than Coding

Generating code is a small part (10-15%) of the job



Coding isn't dead. *It leveled up.*

The floor rose, the job moved up the stack.
Java sits exactly where enterprise AI has to run.